Remark 2.15. Lemma 2.14 gives an alternative definition of fractional ideals that can be extended to domains that are not necessarily noetherian; they are A-submodules I of K for which there exists a nonzero $r \in A$ such that $rI \subseteq A$. When A is noetherian this coincides with our definition above.

Corollary 2.16. *Every fractional ideal of A can be written in the form $\frac{1}{a}I$, for some nonzero $a \in A$ and ideal I.*

Definition 2.17. A fractional ideal of A is *principal* if it is generated by one element, that is, it has the form xA for some $x \in K$. We will also use the notation $(x) := xA$ to denote the principal fractional ideal generated by $x \in K$.

As with ideals, we can add and multiply fractional ideals:

$$I + J := (i + j : i \in I, j \in J), \qquad IJ := (ij : i \in I, j \in J).$$

Here the notation (S) means the A-module generated by $S \subseteq K$. As with ideals, we actually have $I + J = \{i + j : i \in I, j \in J\}$, but the ideal IJ is typically not the same as set $\{ij : i \in I, j \in J\}$, it consists of all finite sums of elements in this set. We also have a new operation, corresponding to division. For any fractional ideals I, J with J nonzero, the set

$$I \div J := \{x \in K : xJ \subseteq I\}$$

is the quotient of I by J. Some texts denote this $(I : J)$ and call it a *colon ideal*. Note that $I \div J$ is **not** a quotient of A-modules (for example, $\mathbb{Z} \div \mathbb{Z} = \mathbb{Z}$ but $\mathbb{Z}/\mathbb{Z} = \{0\}$).

We do not assume $I \subseteq J$ (or $J \subseteq I$), the definition makes sense for any fractional ideals I and J with J nonzero.[2] If $I = (x)$ and $J = (y)$ are principal fractional ideals then $(I \div J) = (x/y)$, so colon ideals can be viewed as a generalization of division in $K^\times$.

Lemma 2.18. *Let I and J be fractional ideals of a noetherian domain A with J nonzero. Then $(I \div J)$ is a fractional ideal of A.*

Proof. It is clear from the definition that $I \div J$ is closed under addition and multiplication by elements of A (since I is), so $I \div J$ is an A-module of the fraction field of A. To show that $I \div J$ is finitely generated, we first suppose that $I, J \subseteq A$ are ideals. For any nonzero $j \in J \subseteq A$ we have $j(I \div J) \subseteq I \subseteq A$, so $(I \div J)$ is finitely generated, by Lemma 2.14. For the general case, choose a and b so that $aI \subseteq A$ and $bJ \subseteq A$ via Lemma 2.14. Then $I \div J = abI \div abJ$ with $abI, abJ \subseteq A$, which we have already shown is finitely generated. $\square$

Definition 2.19. A fractional ideal I is *invertible* if $IJ = A$ for some fractional ideal J.

Inverses are unique when they exist: if $IJ = A = IJ'$ then $J = JA = JIJ' = AJ' = J'$. We may use I^{-1} to denote the inverse of a fractional ideal I when it exists.

Lemma 2.20. *A fractional ideal I of A is invertible if and only if $I(A \div I) = A$ (in which case $A \div I$ is its inverse).*

Before proving the lemma, note that $I(A \div I) \subseteq A$ always holds, since for $y \in I$ and $x \in A \div I$ we have $xy \in xI \subseteq A$, by the definition of $A \div I$. The lemma states that this inclusion is an equality precisely when I is invertible.

[2] The definition still makes sense when J is the zero ideal, but $I \div (0) = K$ will typically not be finitely generated as an A-module, hence not a fractional ideal.